

Software Engineering Intro

Condensed Study Notes

Generated: 2026-05-07 06:44 UTC

Course Contents

This course covers several key areas of software development:

- **Introduction; Software Design:** This encompasses software architecture, design patterns, Object-Oriented (OO) analysis and design, and designing for re-use.
- **Using APIs:** Focuses on API programming, class browsers, and related tools, as well as component-based computing.
- **Software Tools and Environment:** Includes tools for requirements analysis and design modeling, testing tools, and mechanisms for tool integration.

The Term Software

Software is a collection of instructions that guides a system to process inputs and generate desired outputs. These outputs are defined by user-specified functions and performance requirements.

Software also encompasses supporting documentation, such as manuals, which aid users in comprehending the software system.

Today's software is comprised of

Software consists of several key components:

- **Instructions (Computer Programs):** These are the coded commands that, when executed, deliver the intended functions and performance.
- **Data Structures:** These are arrangements that allow programs to efficiently manipulate information.
- **Documents:** These describe how the software operates and is used. Optional features may be based on customer choices.

Software development also considers the **platform**, which includes the hardware and operating system requirements for it to run. Software is therefore described by its capabilities and its platform specifications.

Additional elements related to software include:

- **Source Code:** Programs written in a specific programming language (e.g., Java, Python).
- **Executable:** Compiled code that runs independently of the source code and can be shared.
- **Design Documents:** Various designs like input/output designs, UML diagrams, and architecture blueprints.
- **Operations:** The expected functions of the software, which can be programmer-centered, generic, or customized.
- **System Manuals:** Guides for users and technical staff, such as user guides and technical manuals.

- **Installation:** The process of making the software functional on a user's system, dependent on hardware and OS.
- **Implementation:** Integrating the installed software into the organization's real-life operations without disrupting existing systems, using techniques like direct, pilot, phased, or parallel methods.

The Term Engineering

Engineering is the profession and activity of designing structures like roads and bridges, or machinery and equipment, based on knowledge.

An engineering process involves achieving specific goals within associated constraints. Engineers control processes using measurement, action, and by acquiring needs and resources to produce products or provide services.

More broadly, engineering means to 'make, achieve or get through contrivance or guile.'

- **Contrivance:** A devised machine or specialized mechanical device for a particular function.
- **Guile:** Cunning or deceit used to find solutions, particularly for non-mechanical problems.

Fundamentally, engineering is an action word, a verb, representing a problem-solving approach.

Source Code vs. Executable in Visual Basic

This practical note focuses on distinguishing between source code and executable files specifically within the Visual Basic environment.

Source code is the human-readable set of instructions written by a programmer. An executable file is the machine-readable version of that source code, ready to be run by a computer. Visual Basic is the tool used to create and manage these files.

The Term Software Engineering

Software Engineering applies engineering principles to software development. It focuses on achieving best practices and systematic processes for building software, especially for larger and complex systems.

Key aspects include:

- Applying scientific knowledge to create cost-effective solutions for practical problems.
- Analyzing user needs to design, construct, and test applications that satisfy those needs.
- Using a systematic, disciplined, and quantifiable approach to software development, operation, and maintenance.

Computer Science VS Software Engineering

Computer Science focuses on the theoretical and mathematical aspects of how computers work.

Software Engineering is the study of building software systems, encompassing project management, architecture, risk management, quality assurance, and testing.

Both fields teach programming fundamentals, allowing either to lead to a software developer role.

Choose Computer Science if you prefer math and logic, or wish to specialize in areas like AI, machine learning, security, or graphics.

Choose Software Engineering for a hands-on approach to understanding the complete software development and maintenance lifecycle.

OR

Software Engineering applies scientific methods and knowledge to create cost-effective solutions for software design, construction, operation, and maintenance.

Its goal is to produce fault-free software that meets user needs, delivered on time and within budget.

The term also signifies a movement towards more systematic software development through various methods and techniques.

The key differences are

Computer Science offers more elective choices in advanced topics, including security, software engineering fundamentals, computer vision, machine learning, and database management.

Software Engineering requires more foundational courses in software testing, design, and software requirements specification.

Importance of Software

Software is a driving force in modern society.

Key areas of impact:

- **Business:** It fuels decision-making.
- **Science & Engineering:** It forms the basis for investigation and problem-solving.
- **Embedded Systems:** It is integral to transportation, medical devices, telecommunications, military systems, industrial processes, entertainment, and office products.
- **Societal Pervasiveness:** Software affects nearly every aspect of life, commerce, culture, and daily activities.

As software's importance grows, the software community focuses on developing technologies to build high-quality programs more easily, quickly, and affordably.